

CRUD API (Project)

Date:27/08/25

create a CRUD API,to store,list and delete movie table.
use code first approach to crete the table
also call these apis via ts

[FlimController.cs](#):

```
using Crudoperations.Context;
using Crudoperations.Models;
using Microsoft.AspNetCore.Authorization;
using Microsoft.AspNetCore.Mvc;
using System.Linq.Expressions;

namespace Crudoperations.Controllers
{
    [ApiController]
    [Route("[controller]")]
    public class FlimController : Controller
    {
        appDbContext appDbContext;
        public FlimController(appDbContext ctxt)
        {
            appDbContext = ctxt;
        }
        [Authorize(Roles = "Admin")]
        [HttpPost("addFlim")]
        public IActionResult addFlim(FlimDetails flimDetails)
        {
            appDbContext.Flms.Add(flimDetails);
            appDbContext.SaveChanges();
            return Ok("flim added");
        }
        [HttpDelete("deleteFlim")]
        public IActionResult deleteFlim(int id)
        {
            if(id<0)
            {
                return Ok("only pos num allowed");
            }
            var flimToDelete = appDbContext.Flms.FirstOrDefault(x => x.FlimId == id);
```

```

        if (flimtoDelete==null)
        {
            return Ok("flim not found");
        }
        appDbContext.Flims.Remove(flimtoDelete);
        appDbContext.SaveChanges();
        return Ok("deleted");
    }
    [Authorize(Roles = "User")]
    [HttpGet("fetchall")]
    public IActionResult fetchall()
    {
        var fetchall = appDbContext.Flims.ToList();
        return Ok(fetchall);
    }

    [HttpDelete("deleteFlimbyname")]
    public IActionResult deleteFlimbyname(string flimname)
    {
        var flimtoDelete = appDbContext.Flims.FirstOrDefault(x => x.FlimName == flimname);

        appDbContext.Flims.Remove(flimtoDelete);
        appDbContext.SaveChanges();
        return Ok("deleted");
    }
}
}

```

UserController.cs:

```

using Crudoperations.Context;
using Crudoperations.Models;
using Crudoperations.Security;
using Microsoft.AspNetCore.Authorization;
using Microsoft.AspNetCore.Identity.Data;
using Microsoft.AspNetCore.Mvc;
using System.Security.Claims;

namespace Crudoperations.Controllers
{
    public record LoginRequest(string name, string Password);
    [ApiController]
    [Route("[controller]")]

```

```

public class UserFlimController : Controller
{
    appDbContext appDbContext;
    Microsoft.Extensions.Options.IOptions<JWTOptions> jwtOptions;
    public UserFlimController(appDbContext ctxt,
Microsoft.Extensions.Options.IOptions<JWTOptions> _jwtOptions)
    {
        appDbContext = ctxt;
        jwtOptions = _jwtOptions;
    }

    [HttpPost("addFlimUser")]
    public IActionResult addFlimUser(UsersFlim usersFlim)
    {
        appDbContext.UsersFlim.Add(usersFlim);
        appDbContext.SaveChanges();
        return Ok("flim added");
    }

    const string ADMIN = "admin";
    const string USER= "user";
    [HttpPost("Login")]
    public IActionResult Login(LoginRequest loginReq)
    {
        var name = loginReq.name;
        var Password = (loginReq.Password);
        var logged_in_user = appDbContext.UsersFlim.Where(x => x.Name == name &&
x.Password == Password).FirstOrDefault();
        if (logged_in_user == null)
        {
            return Unauthorized("sry unauthorized");
        }
        else
        {
            var claims = new List<Claim>
            {
                new Claim(ClaimTypes.Name,loginReq.name),
                //new Claim(ClaimTypes.Role,"Admin")
            };
            if (logged_in_user.Role.Trim().ToLower().Contains(ADMIN) ||
logged_in_user.Role.Trim().ToLower().Contains(USER))
            {
                claims.Add(new Claim(ClaimTypes.Role, "Admin"));
                claims.Add(new Claim(ClaimTypes.Role, "User"));
            }
        }
    }
}

```

```

    }

    var token = JwtService.CreateJWTToken(jwtOptions.Value, claims);
    return Ok(token);
}
}
}
}

```

[FlimDetails.cs:](#)

```

using System.ComponentModel.DataAnnotations;

namespace Crudoperations.Models
{
    public class FlimDetails
    {
        [Key]
        public int FlimId { get; set; }
        public string FlimName { get; set; }
        public DateOnly ReleaseDate { get; set; }
    }
}

```

[UserFlim.cs:](#)

```

namespace Crudoperations.Models
{
    public class UsersFlim
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public string Password { get; set; }
        public string Role { get; set; }
    }
}

```

[appDbContext.cs:](#)

```

using Crudoperations.Models;
using Microsoft.EntityFrameworkCore;

namespace Crudoperations.Context

```

```

{
  public class appDbContext:DbContext
  {
    public appDbContext(DbContextOptions options) : base(options)
    {
    }
    public DbSet<FlimDetails> Flims{ get; set; }
    public DbSet<UsersFlim> UsersFlim { get; set; }
  }
}

```

Typescript:

```

// let nam:string="nancy";
// console.log(nam);
interface FlimDetails{
  flimId: number,
  flimName:string,
  releaseDate:Date
}
async function fetchFlimPost(token:string)
{
  const response = await fetch('http://localhost:5095/Flim/fetchall',{method:'GET',
    headers:{
      'Content-Type': 'application/json',
      'Authorization': 'Bearer '+token
    }
  });
  if(!response.ok)
  {
    console.log("error occured");
  }
  const all_flims:FlimDetails=await response.json();
  console.log(all_flims);
}
//
fetchFlimPost(eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJodHRwOi8vc2NoZW1hcy54bWxzbnZlbnZy93cy8yMDA1LzA1L2lkZW50aXR5L2NsYWltcy9uYW11IjoibmlybWFsIiwiaHR0cDovL3NjaGVtYXMubWljcm9zb2Z0LmNvbS93cy8yMDA4LzA2L2lkZW50aXR5L2NsYWltcy9yb2x1IjpbIkFkbWluIiwiaXN1ciJdLCJleHAiOjE3NTYzMTU3MDcsIm1zcyI6I

```

```
mh0dHBz0i8vd3d3LnBheW9kYS5jb20iLCJhdWQiOiJzdHVkZW50c19DU2hhcnAifQ.5NZJ  
wmPIId0lVYDQ1VJM2-q6ipSs-z6cW4bCJzWrCn3c);
```

```
interface _newFlim{  
  flimId: number,  
  flimName: string,  
  releaseDate: string  
}  
const newFlim: _newFlim={  
  flimId: 0,  
  flimName: "Coolie",  
  releaseDate: "2025-08-14"  
}  
async function AddNewBook(token:string)  
{  
  const response = await fetch(`http://localhost:5095/Flim/addFlim`,{method:'POST',  
    body:JSON.stringify(newFlim),  
    headers:{  
      'Content-Type': 'application/json',  
      'Authorization': 'Bearer '+token  
    }  
  });  
  if(!response.ok)  
  {  
    console.log("error occured");  
  }  
  
  console.log(sucessfully added and status ${response.status});  
  
}
```

```
//
```

```
AddNewBook(eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJodHRwOi8vc2NoZW1hcy5  
4bWxzbn2FwLm9yZy93cy8yMDA1LzA1L2lkZW50aXR5L2NsYWltcy9uYW1lIjoibWFyaWEiL  
CJodHRwOi8vc2NoZW1hcy5taWNYb3NvZnQuY29tL3dzLzIwMDgvMDYvaWRlbnRpdHkvY2x  
haW1zL3JvbGUiOiJsiQWRtaW4iLCJvc2VyIl0sImV4cCI6MTc1NjM3MTIyNCwiaXNzIjoia  
HR0cHM6Ly93d3cucGF5b2RhLmNvbSI6ImF1ZCI6InN0dWRlbnRzX0NTaGFycCJ9.Wsrhn7  
wKNokOdTvzeSxwB_cBtNa-ZRupXqdDhoZJjs8);
```

```
async function delFlim(id:number)  
{
```

```

    const response=await
    fetch(`http://localhost:5095/Flim/deleteFlim?id=${id}`, {method: `DELETE`,
      headers:{
        'Content-Type': 'application/json',
      }
    });
    if(!response.ok)
    {
      console.log("error occured");
    }
    else
    {
      console.log(delted sucessfully :${id});
    }
  }
  // delFlim(11);
  async function delFlimbyname(flimname:string)
  {
    const response=await
    fetch(`http://localhost:5095/Flim/deleteFlimbyname?flimname=${flimname}`, {method: `DELETE`,
      headers:{
        'Content-Type': 'application/json',
      }
    });
    if(!response.ok)
    {
      console.log("error occured");
    }
    else
    {
      console.log(delted sucessfully :${flimname});
    }
  }
  delFlimbyname("Jailer")

```

SQLQuery19.sql -...YODA\nancy.m (57)) SQLQuery18.s

```
SELECT TOP (1000) [FlimId]
      ,[FlimName]
      ,[ReleaseDate]
FROM [demo_nancy].[dbo].[Flims]
```

100 %



Results



Messages

	FlimId	FlimName	ReleaseDate
1	2	Vidaamuyarchi	2025-02-06
2	3	Asthram	2025-03-21

SQLQuery20.sql -...ODA\nancy.m (103)) -p X SQLQuery19.sql -...YO

```
SELECT TOP (1000) [Id]
, [Name]
, [Password]
, [Role]
FROM [demo_nancy].[dbo].[UsersFlim]
```

100 %

Results Messages

	Id	Name	Password	Role
1	1	maria	maria	admin
2	2	nimal	nimal	user